the band-pass is better, the peak is not as sharp as the notch filter from which it was derived. These mediocre results are especially disappointing in comparison to the excellent performance seen in Chapter 14. Why the difference? The answer lies in something that is often forgotten in filter design: the *phase response*.

To illustrate how phase is the culprit, consider a system called the **Hilbert transformer**. The Hilbert transformer is not a specific device, but any system that has the frequency response: Magnitude = 1 and phase = 90 degrees, for all frequencies. This means that any sinusoid passing through a Hilbert transformer will be unaffected in amplitude, but changed in phase by one-quarter of a cycle. Hilbert transformers can be analog or discrete (that is, hardware or software), and are commonly used in communications for various modulation and demodulation techniques.

Now, suppose we spectrally invert the Hilbert transformer by subtracting its output from the original signal. Looking only at the *magnitude* of the frequency responses, we would conclude that the entire system would have an output of *zero*. That is, the magnitude of the Hilbert transformer's output is identical to the magnitude of the original signal, and the two will cancel. This, of course, is completely incorrect. Two sinusoids will exactly cancel only if they have the same magnitude *and* phase. In reality, the frequency response of this composite system has a magnitude of $\sqrt{2}$, and a phase shift of -45 degrees. Rather than being zero (our naive guess), the output is *larger* in amplitude than the input!

Spectral inversion works well in Chapter 14 because of the specific kind of filter used: *zero phase*. That is, the filter kernels have a left-right symmetry. When there is no phase shift introduced by a system, the subtraction of the output from the input is dictated solely by the magnitudes. Since recursive filters are plagued with phase shift, spectral inversion generally produces unsatisfactory filters.

Gain Changes

Suppose we have a recursive filter and need to modify the recursion coefficients such that the output signal is changed in amplitude. This might be needed, for example, to insure that a filter has unity gain in the passband. The method to achieve this is very simple: multiply the "a" coefficients by whatever factor we want the gain to change by, and leave the "b" coefficients alone.

Before adjusting the gain, we would probably like to know its current value. Since the gain must be specified at a frequency in the *passband*, the procedure depends on the type of filter being used. Low-pass filters have their gain measured at a frequency of *zero*, while high-pass filters use a frequency of 0.5, the maximum frequency allowable. It is quite simple to derive expressions for the gain at both these special frequencies. Here's how it is done.